

Read each of the questions carefully and answer showing all steps of your solution

Final Answers with no clear solution get ZERO

All the computations must be rounded to exactly four decimal places.

Q1- Find the root of the function: $f(x) = 2x \cos(x) - \sin(x) + 1$

Use the Newton Raphson Method. Perform three iterations to find the root, assume $x_0 = 1$ (15 marks)

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	1	$2x \cos(x) - \sin(x) + 1$	①	2.0844
1	2.0844	①	②	1.6188
2	1.6188	②	③	1.5718
3	1.5718	③	④	1.5708

$$x_{i+1} = \frac{f(x)}{f'(x)}$$

$$f(x) = 2x \cos(x) - \sin(x) + 1$$

$$① \quad f'(x) = -2x \sin(x) + 2 \cos(x) - \cos(x)$$

$$② \quad f''(x) = -2 \cos(x) + 2 \sin(x) - \sin(x)$$

$$③ \quad f'''(x) = +2 \sin(x) + 2 \cos(x) - \cos(x)$$

$$④ \quad f^{(4)}(x) = 2x \cos(x) + 2 \sin(x) + \sin(x)$$

no need

$$h=0 \Rightarrow x_1 = \frac{f(x)}{f'(x)} = 2.0844$$

$$h=1 \Rightarrow x_2 = \frac{f'(x)}{f''(x)} = 1.6188$$

$$h=2 \Rightarrow x_3 = \frac{f''(x)}{f'''(x)} = 1.5718$$

Read each of the questions carefully and answer showing all steps of your solution

Final Answers with no clear solution get ZERO

All the computations must be rounded to exactly four decimal places

Q2. Use the Golden Section Search to find the minimum of the function.

$f(x) = 5e^x + \cos(x)$. Assume that $x_l = -4.5$, $x_u = -2$. Perform 4 iterations.

(15 marks)

n	x_l	x_u	d	x_1	x_2	$f(x_1)$	$f(x_2)$
1	-4.5	-2	$\frac{55-1}{2}$	$d(-4.5, -2) = -4.017$	$d(-4.5, -2) = -1.54$	-0.552	2.665
2	-4.017	-1.54	$\frac{55-1}{2}$	$d(-4.017, -1.54) = -3.434$	$d(-4.017, -1.54) = -1.5308$	-0.556	1.1026
3	-3.434	-1.5308	$\frac{55-1}{2}$	$d = -3.0684$	$d = -1.17$	-0.796	1.234
4	-3.0684	-1.17	$\frac{55-1}{2}$	$d = -3.0684$	$d = -1.17$	-0.764	1.9419

$$f(x_1) = f(-4.5) = 5e^{-4.5} + \cos(-4.5) = -0.552$$

$$f(x_2) = f(-1.54) = 5e^{-1.54} + \cos(-1.54) = 2.665$$

$$f(x_1) = f(-3.434) = 5e^{-3.434} + \cos(-3.434) = -0.556$$

$$f(x_2) = f(-1.5308) = 5e^{-1.5308} + \cos(-1.5308) = 1.1026$$

$$f(x_1) = f(-3.0684) = 5e^{-3.0684} + \cos(-3.0684) = -0.796$$

$$f(x_2) = f(-1.17) = 5e^{-1.17} + \cos(-1.17) = 1.234$$

$$f(x_1) = f(-3.0684) = 5e^{-3.0684} + \cos(-3.0684) = -0.764$$

$$f(x_2) > f(x_1)$$